Let $R=\{(x,y):x,y\in Z, x^2+3y^2\le 8\}$ is a relation on the set of integers Z, then the domain of R^{-1},

A. $\{-1,0,1\}$

B. $\{-2,-1,1,2\}$

C. $\{-2,-1,0,1,2\}$

D. $\{0,1\}$

[JEE 2020]

Consider the relation,

$$R = \{(x, y) : x, y \in Z, x^2 + 3y^2 \le 8\}$$

Thus, if $(x, y) \in Z$ then $x^2 + 3y^2 \le 8$ that is $x^2 \le 8 - 3y^2$

As x is an integer so, $8 - 3y^2 \ge 0$ that is,

$$3y^2 \le 8 \Rightarrow y^2 \le \frac{8}{3} = 2.7 \Rightarrow -1.6 \le y \le 1.6 \Rightarrow y = -1, 0, 1$$

Hence the possible values of y are $y = 0, 1$

For $y = -1$, $x^2 \le 5$ so, $x = 0, 1, 2$

For $y = 0$, $x^2 \le 8$ so, $x = 0, 1, 2$

For $y = 1$, $x^2 \le 5$ so, $x = 0, 1, 2$

Thus,

$$R = \left\{ \begin{matrix} (0,-1),(1,-1),(2,-1),(0,0),(1,0),(2,0), \\ (0,1),(1,1),(2,1) \end{matrix} \right\}$$

That is the domain of the relation R is $\{0, 1, 2\}$ and the range of the relation R is $\{-1, 0, 1\}$ hence, the domain of the relation R^{-1} is $\{-1, 0, 1\}$

Hence, the **OPTION A** is the correct.